

CSET218 – BASIC FRONTEND DEVELOPMENT

Course Type-Specialization Core L-T-P Format 6-0-4 Credits–4

The course covers **Basic Front-End Development, Client-server Architecture, Frontend Programming Tools, Frameworks, and Libraries**. This includes basics of **HTML, CSS and JS**. We will be doing 3 Basic Projects along with that.

COURSE-SPECIFIC LEARNING OUTCOMES

CO1: To understand the Client-Server architecture and the basics of client-side programming languages and frameworks.

CO2: To examine of client-side languages: HTML, CSS, and JavaScript and Bootstrap.

CO3: To implement client-side application development.

DETAILED SYLLABUS

Module 1 (W1, W2, W3, W4) Introduction to HTML, Character Set and Encoding, Document Type, Elements, Attributes, Comments, Display, HTML Elements, Body Elements, Heading Element, Paragraph Element, Link Element, List Element, Table Element, Image Element, Figure Element, Picture Element, Icon Element, Video Element, Audio Element, iFrame Element, From Element, Block Quote Element, Marque Element, Head Elements, Title Element, Style Element, Script Element, Meta Element, Base Element and HTML Entities.

Module 2 (W5, W8, W9, W10) Introduction to CSS, Selectors-I, Adding CSS, Comments, Units, Selectors-II, Variables, Counters, Specificity, Important, Maths Functions, Properties in CSS, Colours, Border, Margin, Padding, Height Weight, Outline, Background, Font, Gradient, Shadow, Tooltip, Element Styling, Text, Link, List, Table, Form, Images, Button, Layout in CSS, Display, Position, z-Index, Overflow, Align, Float, Clear, Flexbox, Grid, Responsiveness, Media Query and View Port.

Module 3 (W11, W12) What is Bootstrap? Creating Our Website, Bootstrap Components, Adding Navbar to Website, Adding Breadcrumb component in Website, Alerts Component in Bootstrap, Badge Component, Buttons in Bootstrap, Button group in Bootstrap, Cards in Bootstrap, Carousels in Bootstrap, Grid System in Bootstrap, Website Project (Blog Site), Adding Navigation bar to Website, Adding Carousel to Website, Adding Custom Components to Website, Creating "About" webpage, Adding Footer in website, Creating "Contact" Webpage, Adding Modals (Login & Signup) to Website, Hosting for Website, SEO of Website, Dynamic Website Free Course.

Module 4 (W17, W18, W19, W20, W21, W22, W23, W24) Introduction to JavaScript, Variables, Datatypes and Operations, Numbers, Control Statements, Functions, Strings, Arrays, Dates, Iterables, Browser Object Model, Window Object, Screen Object, Location, Object, History Object, Navigator Object, Document Object Model, Objects in DOM, Document Object, Element Object, Event Object, Node Object, HTMLCollection Object, NodeList Object, Object Oriented Programming, Objects and Classes.

STUDIO WORK / PROJECTS

Module 1 (W1, W2, W3, W4) N/A

Module 2 (W5, W8, W9, W10) N/A

Module 3 (W11, W12) Project-1(HTML, CSS, Lensshub Basics), Project-2(HTML, CSS, My Portfolio).

Module 4 (W17, W18, W19, W20, W21, W22, W23, W24) Project-3(HTML, CSS, JS, Password Generator and To-Do List).

EXAMINATIONS

Mid-Term and Syllabus (W6, W7) Module-1, Module-2.

End-Term and Syllabus (W13, W14, W15, W16) Complete Syllabus.

